ComNet assgn-01

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1. Given the frequencies listed below, calculate the corresponding periods.
   a. 24 Hz.
   b. 8 MHz
2. Given the following periods, calculate the corresponding frequencies.
   a. 5 seconds
   b. 12 micro-seconds
3. If the bandwidth of the channel is 5 Kbps. How long does it take to send a frame of 100,000 bitsout of this device?
4. A line has a signal-to-noise ratio of 1000 and a bandwidth of 4000 KHz. What is the maximum data rate supported by this line?
5. A signal with 200 milliwatts power passes through 10 devices, each with an average noise of 2 microwatts. What is the SNR? What is the SNRdB?

Answers will be below

1. $Period = \frac{1}{Frequency}$
   1. $\frac{1}{24} = 0.042\ second$
   2. $\frac{1}{8 \times 10^6} = 0.125\ \mu S$
2. $frequency = \frac{1}{period}$
   1. $\frac{1}{5} = 0.2\ Hz$
   2. $12\mu S = 12 \times 10^{-6}\ second$
      $\frac{1}{12 \times 10^{-6}} = 83333.33\ Hz$
3. Kpbs means kilo bits per sec, which means, 1 second 1000 bits are sent 5 Kpbs = 5000 bits are sent per 1 second
   $\frac{100{,}000}{5{,}000} = 20\ seconds$
4. Noisy channel data rate limit, Shannon capacity

$$bit\ rate\ capacity = Bandwidth \times log_2(1 + SNR)$$

   $bit\ rate = 4000 \times 10^3 \times log_2(1 + 1000) = 39.87\ Mbps$
5.
   1. $SNR = \frac{200 \times 10^{-3}}{10 \times 2 \times 10^{-6}} = 10,000$
   2. $SNRdB = 10 \times log_{10}(SNR)$, $SNRdB = 10 \times log_{10}(10,000) = 40dB$